



HIBERNATE

Hibernate is an ORM (Object-Relational Mapping) framework for Java that maps Java objects to relational database tables and handles database operations automatically.

Advantages:

- Reduces boilerplate JDBC code.
- Database-independent (uses HQL).

Hibernate configuration file: (hibernate.config.xml or hibernate.properties)

- This file contains database-related configuration and session-related configuration.

JDBC - Java Database Connectivity
Advantages of Hibernate over JDBC

<u>JDBC</u>	<u>Hibernate</u>
Manual SQL	Automatic SQL generation
More code	Less code
No caching	Supports caching
DB dependent	DB independent

- Lazy loading: Data is loaded only when required, not upfront.
It is mainly used to improve application performance by loading child objects only when the parent object is accessed.
- Eager loading: Data is loaded upfront by loading child objects along with the parent object.

Relationship b/w SessionFactory & Session

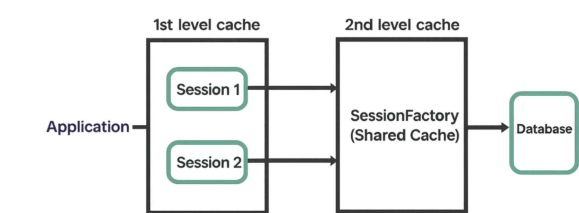
Application Startup	v +-----+	SessionFactory	-----	Heavy-weight	Thread-safe	One per DB	+-----+
	openSession()	v +-----+	Session	-----	Lightweight	Not thread-safe	On e per request
	+-----+	-----	v v save()	update()	delete()	v Database	

Important interfaces in Hibernate

- SessionFactory: A heavy object created once per application; used to create Session objects.
 - Session: Maintains the connection between Java objects and the database.
 - Transaction: Represents a unit of work; handles commit and rollback.
 - Query: Used to create and execute HQL or SQL queries.
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- Diff b/w get() & load() :
 - get() hits db immediately & returns null if not found.
 - load() lazy loading & throws exception if not found.
 - Cache: A temporary memory area where frequently accessed data is stored so repeated queries can be served faster without hitting the database. This improves Hibernate performance by reducing the number of times it needs to communicate with the DB.
 - **Hibernate uses 2 levels of caching :**

First Level Cache(Session level)	Second Level Cache(Session Factory level)
This Cache is local to the session object & can't be shared with different sessions.	This Cache is maintained at <u>SessionFactory</u> level & main benefit from this cache is different sessions can share cached entities to avoid <u>db</u> queries.
Available only until session is open once closed, cache will be destroyed.	Available throughout the application. It is only destroyed & recreated when application is restarted.

- Entity lookup order :



- If Session requests an entity, hibernate follows this order :

